ARCHIT KUMAR

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EDUCATION

Vellore Institute of Technology Sep 2022 – May 2026
Bachelor of Technology in Computer Science, GPA: 8.45/10 Vellore, Tamil Nadu

PROFESSIONAL SUMMARY

Fresher Software Engineer (VIT, GPA: 8.45) with hands-on experience building backend services, REST APIs, and SQL-backed applications using Java, Spring Boot, PostgreSQL, and MySQL. Comfortable across the software development lifecycle, including debugging, unit testing, Git, and Agile team delivery. Seeking an entry-level role in backend or full-stack development.

TECHNICAL SKILLS

Programming Languages: Java, C++, Python, JavaScript, SQL
Backend & Distributed Systems: Spring Boot, FastAPI, REST APIs, Microservices, Spring Data JPA
Databases & Messaging: PostgreSQL, MySQL, Redis, Apache Kafka, pgvector
Frontend: HTML5, CSS3, JavaScript, React.js, Next.js, Tailwind CSS
Tools & Platforms: Docker, Git, GitHub, Postman, AWS, k6, Celery, Socket.io, Prometheus
CS Fundamentals: Object-Oriented Programming (OOP), Data Structures & Algorithms, SDLC, Debugging, DBMS, Operating Systems, Computer Networks, Unit Testing, Agile

PROFESSIONAL EXPERIENCE

SDE Intern — Agent Evaluation & Benchmark Engineering May 2026 – Oct 2026
Xelron.ai *Remote*

- Authored and validated hard, verifiable benchmark tasks for frontier coding agents (GPT-5.5/Codex, Claude Opus) across ML, security, systems administration, and build-tooling domains — task specifications, deterministic behavioral tests, Dockerized environments, oracle solutions, and anti-cheat safeguards — maintaining an 87% approval rate across 4 project workstreams.
- Promoted to Team Lead within two months, leading a 10-member task-authoring team through technical reviews, agent-failure trajectory analysis, and difficulty calibration to keep submissions within the target solve-rate band.
- Designed rubric-scored, multi-step evaluation tasks spanning coding-agent benchmarks and domain-specific occupational scenarios, building realistic input datasets, gold-standard reference solutions, and weighted scoring rubrics validated against model outputs.
- Used Cursor, Claude, and GPT-based tools daily for repository analysis, debugging, and implementation planning, independently verifying submissions via CI checks, containerized test harnesses, and repeated agent evaluation runs.

GenAI Intern (WatsonX Training Program) May 2025 – July 2025
IBM *Remote*

- Implemented Retrieval-Augmented Generation (RAG) pipelines using WatsonX foundation models for document-based Q&A and summarization.
- Built an AI chatbot on IBM Watson Assistant, applying prompt engineering to design conversational flows and structured prompts for text classification, summarization, and code generation.

KEY PROJECTS

Distributed Inventory Reservation System | Spring Boot, PostgreSQL, Redis, Apache Kafka, React, Docker, k6 | GitHub

- Implemented a concurrency-safe Inventory Reservation System using Spring Boot and PostgreSQL pessimistic locking, ensuring zero-overselling across 100+ simultaneous requests as validated by k6 load testing.
- Architected an event-driven stock-release workflow via Apache Kafka for failed orders and implemented Idempotency-Key support to maintain system consistency and prevent duplicate stock deductions during retries.
- Optimized lookup latency using Redis caching with transaction-aware cache eviction and established a robust testing suite using Testcontainers for automated PostgreSQL integration testing in isolated Docker environments.

Ad-Keyword-Targeting Application | Java, Spring Boot, React, MySQL, MinIO, Docker, JWT Authentication | GitHub

- Developed a full-stack advertisement management platform with JWT-based authentication, secure media uploads via MinIO object storage, and paginated ad discovery APIs supporting keyword search, sorting, and filtering workflows.
- Optimized high-volume ad retrieval by designing a normalized MySQL schema with strategic indexing and caching-aware API patterns, reducing query latency by ~40% during concurrent search operations.
- Designed scalable backend services with Spring Boot, Spring Security, and OpenAPI documentation while containerizing infrastructure using Docker Compose for reproducible local deployment and testing.

Climate Risk Prediction System | FastAPI, React, PostgreSQL, Redis, Celery, PyTorch, XGBoost, Docker | GitHub | Demo

- Architected a distributed climate risk analysis platform using FastAPI, Celery, Redis, and PostgreSQL to process asynchronous satellite-image inference workflows with real-time dashboard updates via Socket.io.
- Developed an ensemble ML pipeline combining Vision Transformers, ResNet feature extraction, and XGBoost classification, achieving ~87% prediction accuracy with SHAP-based explainability for transparent risk assessment.
- Engineered full-stack geospatial analytics features including interactive risk visualization, vector similarity search with pgvector, and LLM-driven report generation using LangChain and Gemini APIs.

LEADERSHIP & ACTIVITIES

Core Member July 2023 – Apr 2024
AI/ML Club *Vellore Institute of Technology*

- Contributed to ML/AI workshops and hackathons; mentored peers on coding challenges.